Referência de VHDL

Bibliotecas:

```
Library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;
```

Entidades:

Declaração:

```
entity EntityName is
port (
   FirstPort : in std_logic,
   SecondPort : out std_logic,
   ThirdPort : inout std_logic_vector(15 downto 0) );
end entity;
```

Architecture:

```
Architecture architecturename of entityname is
   -- signals aqui
   -- componentes aqui:
   --
   -- component componentname is
   --      port(
   --            port declaration here.
   --      );
   -- end component;

begin

end architecture;
```

Assignment:

Normal:

```
   A <= B and C;
```

When:

```
   A <= "0000" when D = '0' else "0001";
```

With seelect:

```
   with a select
   b <= "1000" when "00",
         "0100" when "01",
         "0010" when "10",
         "0001" when others;
```

Concatenação:

```
   vec <= bit1 & bit0;
```

Port Map:

```
   u1: modulo1 port map(
        a => i1,
        b => i2,
        x => open
      );
```

Process:

```
process(-- lista
   clk, reset
)
begin
   -- condições
end process;
```

```
if Boolean_expr_1 then
         sequential_statements;
elsif Boolean_expr_2 then
         sequential_statements;
else
         sequential statements;
end if;
```

```
case sel is
         when choice_1 =>
                  sequential_statements;
         when choice_2 =>
                  sequential_statements;
         . . .
         when others =>
                  sequential_statements;
end case;
```